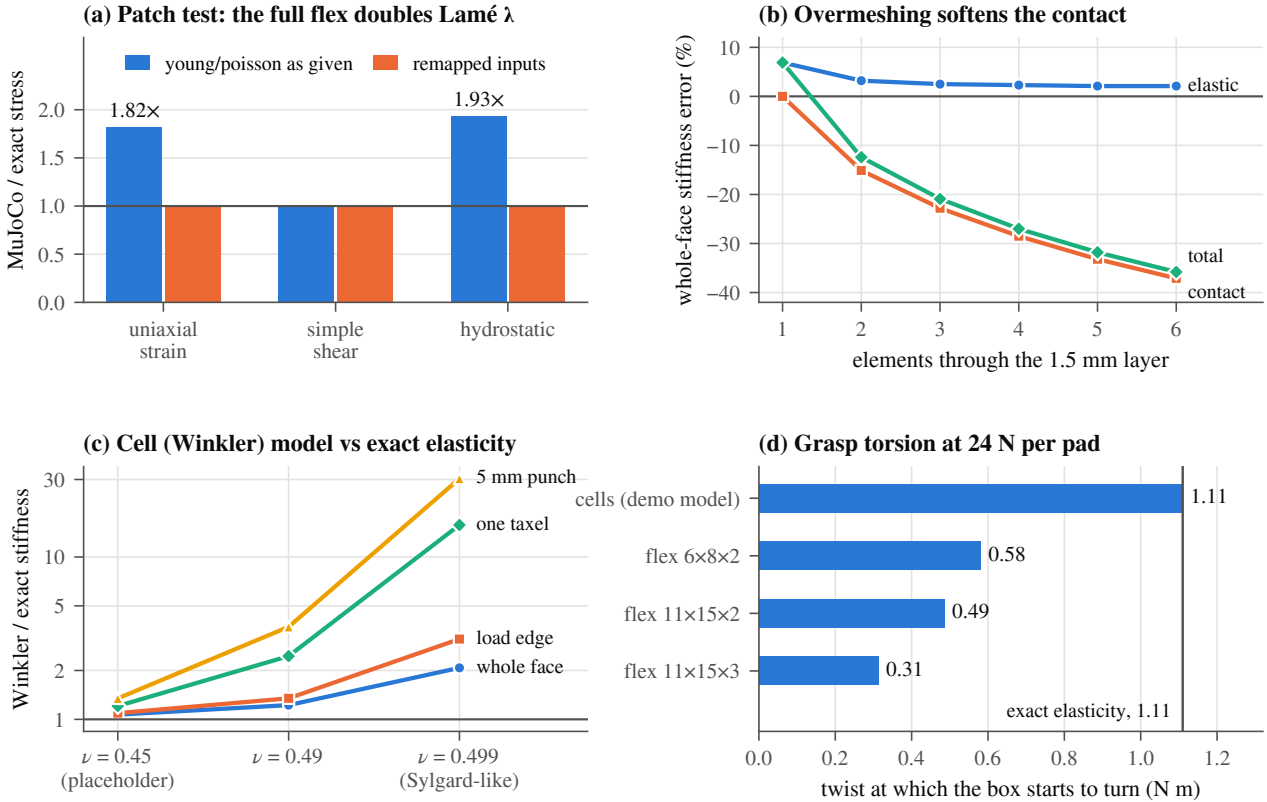
(a) Patch test: the full flex doubles Lamé λ
young/poisson as given
remapped inputs
MuJoCo / exact stress
2.0
1.5
1.0
0.5
0.0
1.82×
1.93×
uniaxial strain
simple shear
hydrostatic
(b) Overmeshing softens the contact
whole-face stiffness error (%)
10
0
−10
−20
−30
−40
elastic
total
contact
1
2
3
4
5
6
elements through the 1.5 mm layer
(c) Cell (Winkler) model vs exact elasticity
Winkler / exact stiffness
30
10
5
2
1
5 mm punch
one taxel
load edge
whole face
ν = 0.45 (placeholder)
ν = 0.49
ν = 0.499 (Sylgard-like)
(d) Grasp torsion at 24 N per pad
cells (demo model)
flex 6×8×2
flex 11×15×2
flex 11×15×3
1.11
0.58
0.49
0.31
exact elasticity, 1.11
0.0
0.2
0.4
0.6
0.8
1.0
1.2
twist at which the box starts to turn (N m)